Tuple Creation

In [2]:
```
t1 = ()
```

In [4]:
```
t2 = (14,56,78)
```

In [6]:
```
t3 = (10.22,45.23,45.67)
```

In [10]:
```
t5 = ('arya',27,(56,78),(390,267))
```

In [12]:
```
t4 = ('one','two','three')
```

In [14]:
```
t6 = (356,'arya',26.789)
```

In [16]:
```
t7 = ('arya',67,[67,890],[567,90],{'shraddha','pranali'},(67,89,34))
```

In [18]:
```
len(t7)
```

Out[18]:
```
6
```

Tuple Indexing

In [21]:
```
t2[0]
```

Out[21]:
```
14
```

In [23]:
```
t4[0]
```

Out[23]:
```
'one'
```

In [25]:
```
t4[0][0]
```

Out[25]:
```
'o'
```

In [27]:
```
t4[-1]
```

Out[27]:
```
'three'
```

Tuple Slicing

In [32]:
```
t = ('shraddha','renuka','nikita','pranali','snehal','nirjala','nidhi','sneha')
```

In [34]:
```
t[0:3]
```

Out[34]:
```
('shraddha', 'renuka', 'nikita')
```

In [36]:
```
t[2:5]
```

Out[36]: ('nikita', 'pranali', 'snehal')

In [38]:
```
t[:3]
```

Out[38]: ('shraddha', 'renuka', 'nikita')

In [40]:
```
t[:2]
```

Out[40]: ('shraddha', 'renuka')

In [42]:
```
t
```

Out[42]:
```
('shraddha',
 'renuka',
 'nikita',
 'pranali',
 'snehal',
 'nirjala',
 'nidhi',
 'sneha')
```

In [44]:
```
t[-3:]
```

Out[44]: ('nirjala', 'nidhi', 'sneha')

In [48]:
```
t[-2:]
```

Out[48]: ('nidhi', 'sneha')

In [50]:
```
t[-1]
```

Out[50]: 'sneha'

In [52]:
```
t[:]
```

Out[52]:
```
('shraddha',
 'renuka',
 'nikita',
 'pranali',
 'snehal',
 'nirjala',
 'nidhi',
 'sneha')
```

Remove and Change item

In [55]:
```
t
```

Out[55]:
```
('shraddha',
 'renuka',
 'nikita',
 'pranali',
 'snehal',
 'nirjala',
 'nidhi',
 'sneha')
```

In [63]:
```
del t[0]  # in tuple we cannot change items
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[63], line 1
----> 1 del t[0]

TypeError: 'tuple' object doesn't support item deletion
```

In [65]:
```
t[0] = 1  # tuples are immutable
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[65], line 1
----> 1 t[0] = 1

TypeError: 'tuple' object does not support item assignment
```

In [67]:
```
del t  # deleting entire tuple object is possible
```

Loop through a tuple

In [72]:
```
t = ('shraddha','renuka','nikita','pranali','snehal','nirjala','nidhi','sneha')
```

In [74]:
```
for i in t:
    print(i)
```

```
shraddha
renuka
nikita
pranali
snehal
nirjala
nidhi
sneha
```

In [76]:
```
for i in enumerate(t):
    print(i)
```

```
(0, 'shraddha')
(1, 'renuka')
(2, 'nikita')
(3, 'pranali')
(4, 'snehal')
(5, 'nirjala')
(6, 'nidhi')
(7, 'sneha')
```

Tuple Membership

In [79]:
```
t
```

Out[79]:
```
('shraddha',
 'renuka',
 'nikita',
 'pranali',
 'snehal',
 'nirjala',
 'nidhi',
 'sneha')
```

In [81]:
```
'shraddha' in t
```

Out[81]:
```
True
```

In [85]:
```
'anjali' in t
```

Out[85]:
```
False
```

In [89]:
```
if 'sneha' in t:
    print('sneha is present in the tuple')
else:
    print('sneha is absent in the tuple')
```

```
sneha is present in the tuple
```

In [91]:
```
if 'aaradhya' in t:
    print('aaradhya is present in the tuple')
else:
    print('aaradhya is absent in the tuple')
```

```
aaradhya is absent in the tuple
```

Index Position

In [94]:
```
t
```

Out[94]:
```
('shraddha',
 'renuka',
 'nikita',
 'pranali',
 'snehal',
 'nirjala',
 'nidhi',
 'sneha')
```

In [96]:
```
t.index('shraddha')
```

Out[96]:
```
0
```

In [98]:
```
t.index('sneha')
```

Out[98]:
```
7
```

In [100…
```
t
```

Out[100…
```
('shraddha',
 'renuka',
 'nikita',
 'pranali',
 'snehal',
 'nirjala',
 'nidhi',
 'sneha')
```

Sorting

In [105…
```
t2 = (34,56,23,67,89,23,56,67)
```

In [107…
```
sorted(t2)   # Returns a new sorted list and dosen't change original tuple
```

Out[107…
```
[23, 23, 34, 56, 56, 67, 67, 89]
```

In [109…
```
sorted(t2, reverse=True) # sort in descending order
```

Out[109…
```
[89, 67, 67, 56, 56, 34, 23, 23]
```

In [ ]: